

1. (5 points) Starting with the Euler's relationship: $e^{j\theta} = \cos \theta + j \sin \theta$ show that $\cos 2x = \cos^2 x - \sin^2 x$

use Euler's formula

$$e^{j\theta} = \cos \theta + j \sin \theta \rightarrow \theta = 2x$$

$$\approx e^{j2x} = (e^{jx})^2 \rightarrow e^{j2x} = (\cos x + j \sin x)^2 = \cos^2 x - \sin^2 x + j(2 \sin x \cos x)$$

$$\cos^2 x + 2j \sin x \cos x + j^2 \sin^2 x \rightarrow (\cos^2 x - \sin^2 x) + j(2 \sin x \cos x)$$

$$\cos 2x + j \sin 2x = (\cos^2 x - \sin^2 x) + j(2 \sin x \cos x)$$

Note that the imaginary parts will go away because of the $A + jB = C + jD$ rule

$$A = C \quad B = D$$

Final Answer

$$\cos 2x = \cos^2 x - \sin^2 x$$

2. (5 Points) Compute the exponential Fourier series of an impulse train $x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$. Now use the computed Fourier series to find the Fourier transform of the impulse train. What can you conclude from the result?

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0) \quad T_0 \rightarrow \omega_0 = \frac{2\pi}{T_0}$$

$$\text{Remember: } X_n = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) e^{-jn\omega_0 t} dt = \frac{1}{T_0} \int \delta(t) e^{-jn\omega_0 t} dt = \frac{1}{T_0}$$

$$x(t) = \frac{1}{T_0} \sum_n e^{jn\omega_0 t}$$

$$e^{j\omega_0 t} \longleftrightarrow 2\pi \delta(\omega - \omega_0)$$

$$X(\omega) = \frac{1}{T_0} \sum 2\pi \delta(\omega - n\omega_0) = \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$$

Final Answer

$$\omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$$

3. (10 Points) Starting with the original sine/cosine Fourier representation,

$$x(t) = a_0 + \sum_{n=0}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

derive the exponential form of the Fourier series

$$x(t) = \sum_{n=-\infty}^{\infty} x_n e^{jn\omega_0 t}$$

with $x_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) e^{-jn\omega_0 t} dt$. For full credit you need to show the derivation of the expression for x_n too. This means, you need to show that $x_n =$

$$\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) e^{-jn\omega_0 t} dt$$

$$\cos n\omega_0 t = \frac{e^{jn\omega_0 t} + e^{-jn\omega_0 t}}{2} \quad \sin n\omega_0 t = \frac{e^{jn\omega_0 t} - e^{-jn\omega_0 t}}{2j}$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} \left[\left(\frac{a_n - jb_n}{2} \right) e^{jn\omega_0 t} + \left(\frac{a_n + jb_n}{2} \right) e^{-jn\omega_0 t} \right]$$

$$\omega \rightarrow T_0 = 2\pi$$

$$\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) e^{-jn\omega_0 t} dt = X_n T_0 \rightarrow X_n = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) e^{-jn\omega_0 t} dt$$

Final Answer

$$X_n = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) e^{-jn\omega_0 t} dt$$

$$\begin{aligned} x(t) &= a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] \\ &= a_0 + \sum_{n=1}^{\infty} \left[\frac{(a_n - jb_n)}{2} e^{jn\omega_0 t} + \frac{(a_n + jb_n)}{2} e^{-jn\omega_0 t} \right] \\ X_n &= \frac{(a_n - jb_n)}{2} \quad X_0 = a_0 \\ X_{-n} &= \frac{(a_n + jb_n)}{2} = X_n^* \\ &= a_0 + \sum_{n=1}^{\infty} [X_n e^{jn\omega_0 t} + X_{-n} e^{-jn\omega_0 t}] \\ &= \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} \end{aligned}$$

4. (5 Points) From the above, derive the Parseval's theorem for the exponential form of the Fourier series.

$$P = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} |x(t)|^2 dt = \frac{1}{T_0} \int \underbrace{x(t) x^*(t)}_{\sum_m x_m^* e^{-jm\omega_0 t}} dt$$

$$P = \frac{1}{T_0} \int \sum_n \sum_m x_n x_m^* e^{j(n-m)\omega_0 t} dt \longrightarrow \sum_n \sum_m x_n x_m^* \left[\frac{1}{T_0} \int e^{j(n-m)\omega_0 t} dt \right]$$

$$x(t) = \begin{cases} 1, & n=m \\ 0, & n \neq m \end{cases}$$

Parseval's theorem $\longrightarrow \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |x_n|^2$

5. (5 Points) Show that the Fourier transform of $\int_{-\infty}^t x(t) dt = \frac{X(\omega)}{j\omega} + \pi X(0) \delta(\omega)$. How do you compute $X(0)$?

$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

$$x(t) * u(t)$$

$$= (x * u)(t) \rightarrow \int x(\tau) u(t - \tau) d\tau \quad \tau < t$$

$$= u(t - \tau) = 1$$

$$Y(\omega) = X(\omega) \cdot U(\omega)$$

$$u(t) \longleftrightarrow \frac{1}{j\omega} + \pi \delta(\omega):$$

$$Y(\omega) = X(\omega) \left[\frac{1}{j\omega} + \pi \delta(\omega) \right] = \frac{X(\omega)}{j\omega} + \pi X(\omega) \delta(\omega)$$

$$Y(\omega) = \frac{X(\omega)}{j\omega} + \pi X(0) \delta(\omega)$$

$$\omega = 0$$

$$X(0) = \int_{-\infty}^{\infty} x(t) dt$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t) \cos(\omega t) dt - j \int_{-\infty}^{\infty} x(t) \sin(\omega t) dt$$

6. (5 Points) Show that the Fourier transform for an even signal $x(t)$ is given by

$$X(\omega) = 2 \int_0^{\infty} x(t) \cos(\omega t) dt$$

$$\text{Euler's formula} = e^{-j\omega t} = \cos(\omega t) - j \sin(\omega t)$$

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t) [\cos(\omega t) - j \sin(\omega t)] dt = \int_{-\infty}^{\infty} x(t) \underbrace{\cos(\omega t)}_{\text{even}} dt - j \int_{-\infty}^{\infty} x(t) \underbrace{\sin(\omega t)}_{\text{odd}} dt \\ I_1 &= \int_{-\infty}^{\infty} x(t) \cos(\omega t) dt & I_2 &= \int_{-\infty}^{\infty} x(t) \sin(\omega t) dt = 0 \\ &= 2 \int_0^{\infty} x(t) \cos(\omega t) dt \end{aligned}$$

$$X(\omega) = I_1 - j I_2 = 2 \int_0^{\infty} x(t) \cos(\omega t) dt - j(0)$$

Final Answer:

$$2 \int_0^{\infty} x(t) \cos(\omega t) dt$$

7. (5 Points) Compute the Fourier transform of this signal: $x(t) = e^{-|t|} \cos(\omega_0 t)$. Use the properties of Fourier transform.

$$X(\omega) \longleftrightarrow x(t) = e^{-|t|} = \int_{-\infty}^0 e^t e^{-j\omega t} dt + \int_0^{\infty} e^{-t} e^{-j\omega t} dt = \frac{1}{1-j\omega} + \frac{1}{1+j\omega} = \frac{2}{1+\omega^2}$$

$$\text{modulation of } \cos(\omega_0 t) = \frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}] = \frac{1}{2} x(t) e^{j\omega_0 t} + \frac{1}{2} x(t) e^{-j\omega_0 t}$$

$$x(t) \cos \omega_0 t \longleftrightarrow \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$$

$$x(t) e^{j\omega_0 t} \longleftrightarrow X(\omega - \omega_0)$$

$$X(\omega) = \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$$

replace every ω with $(\omega - \omega_0)$

$$\frac{1}{2} \cdot \frac{2}{1+(\omega - \omega_0)^2} + \frac{1}{2} \cdot \frac{2}{1+(\omega + \omega_0)^2} \rightarrow \frac{1}{1+(\omega - \omega_0)^2} + \frac{1}{1+(\omega + \omega_0)^2} = X(\omega)$$

Final Answer:

$$X(\omega) = \frac{1}{1+(\omega - \omega_0)^2} + \frac{1}{1+(\omega + \omega_0)^2}$$

8. (10 Points) Compute the Fourier transform of the following signal

$$x(t) = e^{-at} \cos(\omega_0 t) u(t)$$

$$\mathcal{F}[e^{-at} u(t)] = \frac{1}{a+j\omega} \leftarrow \int_0^{\infty} e^{-at} e^{-j\omega t} dt \rightarrow \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$\cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$

$$\frac{1}{2} [e^{-at} e^{j\omega_0 t} u(t) + e^{-at} e^{-j\omega_0 t} u(t)]$$

$$\downarrow$$

$$\frac{1}{a+j(\omega-\omega_0)}$$

$$\downarrow$$

$$\frac{1}{a+j(\omega+\omega_0)}$$

$$\frac{1}{2} \left[\frac{1}{a+j(\omega-\omega_0)} + \frac{1}{a+j(\omega+\omega_0)} \right] \rightarrow \frac{1}{2} \left[\frac{2a + 2j\omega}{a^2 + j2a\omega + j^2\omega^2 + \omega_0^2} \right]$$

$$\mathcal{F}[e^{-at} \cos(\omega_0 t) u(t)] = \frac{a+j\omega}{(a+j\omega)^2 + \omega_0^2}$$